2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

GENERAL PROXIMITY SENSOR MODULE (GPSM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'DTC monitor' tab on the manufacturers approved diagnostic system
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSM's which may be valid for the specific customer complaint and carry out the recommendations as needed

The table below lists all diagnostic trouble codes (DTCs) that could be

logged in the General Proximity Sensor Module (GPSM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.
For additional information, refer to: Parking Aid (413-13 Parking Aid, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B143A-12	Proximity Sensor #1 - Circuit short to battery	▪ Left proximity sensor signal circuit short circuit to power	▪ Refer to the electrical circuit diagrams and test the left proximity sensor signal circuit for short circuit to power
B143A-14	Proximity Sensor #1 - Circuit short to ground or open	▪ Left proximity sensor signal circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and test the left proximity sensor signal circuit for short circuit to ground, open circuit, high resistance
B143A-96	Proximity Sensor #1 - Component internal failure	▪ Left proximity sensor failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new left proximity sensor
B143B-12	Proximity Sensor #2 - Circuit short to battery	▪ Right proximity sensor signal circuit short circuit to power	▪ Refer to the electrical circuit diagrams and test the right proximity sensor signal circuit for short circuit to power
B143B-14	Proximity Sensor #2 - Circuit short to ground or open	▪ Right proximity sensor signal circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and test the right proximity sensor signal circuit for short circuit to ground, open circuit, high resistance
B143B-96	Proximity Sensor #2 - Component internal failure	▪ Right proximity sensor failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new right proximity sensor
B145D-	Proximity	▪ Left proximity	▪ Refer to the electrical circuit

12	Sensor Supply - Circuit #1 - Circuit short to battery	sensor ground circuit short circuit to power	diagrams and test the left proximity sensor ground circuit for short circuit to power
B145D-14	Proximity Sensor Supply - Circuit #1 - Circuit short to ground or open	▪ Left proximity sensor power supply circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and test the left proximity sensor power supply circuit for short circuit to ground, open circuit, high resistance
B145E-12	Proximity Sensor Supply - Circuit #2 - Circuit short to battery	▪ Right proximity sensor ground circuit short circuit to power	▪ Refer to the electrical circuit diagrams and test the right proximity sensor ground circuit for short circuit to power
B145E-14	Proximity Sensor Supply - Circuit #2 - Circuit short to ground or open	▪ Right proximity sensor power supply circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and test the right proximity sensor power supply circuit for short circuit to ground, open circuit, high resistance
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module via the medium speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	▪ Invalid data received from another control module via the high speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	▪ Missing message from another control module via the high speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC

			index
U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-57	Internal Control Module Software Incompatibility - Invalid / incomplete software component	▪ Car configuration file mismatch with vehicle specification	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. Check DTCs in all control modules ▪ In case U0300-00 set in multiple control modules on the chassis high speed CAN bus - Re-configure the central junction box with the latest level software ▪ In case U0300-00 set in wade aid control module only - Re-configure the wade aid control module with the latest level software. If the fault persists, install a new wade aid control module
U2300-54	Central Configuration - Missing calibration	▪ Car configuration file mismatch with vehicle specification	▪ Using the manufacturer approved diagnostic system, check and update the Car Configuration File (CCF) as required. Clear the DTC and retest
U2300-55	Central Configuration - Not configured	▪ Car configuration file mismatch with vehicle specification	▪ Using the manufacturer approved diagnostic system, check and update the Car Configuration File (CCF) as required. Clear the DTC and retest
U2300-56	Central Configuration - Invalid / incomplete configuration	▪ Car configuration file mismatch with vehicle specification	▪ Using the manufacturer approved diagnostic system, check and update the Car Configuration File (CCF) as required. Clear the DTC and retest

U2300-64	Central Configuration - Signal plausibility failure	▪ Car configuration file mismatch with vehicle specification	▪ Using the manufacturer approved diagnostic system, check and update the Car Configuration File (CCF) as required. Clear the DTC and retest
U3000-47	Control Module - Watchdog / safety MicroController failure	▪ General proximity sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a general proximity sensor module
U3000-49	Control Module - Internal electronic failure	▪ General proximity sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new general proximity sensor module
U3000-55	Control Module - Not configured	▪ General proximity sensor module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and test the general proximity sensor module power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
U3000-63	Control Module - Circuit / component protection time-out	▪ General proximity sensor module internal circuit protection routine has deactivated the module following detection of repeated proximity sensor short circuit faults	**NOTE:** This DTC is only logged if the control module has detected persistent proximity sensor short circuit faults. Ensure that all circuit faults have been resolved before proceeding with the reset short protection routine ▪ Check for related general proximity sensor circuit fault DTCs and ensure all proximity

			sensor faults are resolved ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Reset/Clear Specified Function (Reset Short Protection) (0x040E)